



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/21

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45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 A gas is heated. The pressure is kept constant.

Which statement describes the behaviour of the particles in the gas?

- A The particles move faster and become closer together.
- B The particles move faster and become further apart.
- C The particles move more slowly and become closer together.
- D The particles move more slowly and become further apart.

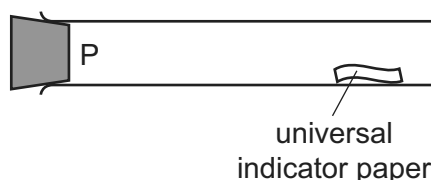
- 2 A mixture of ice and water is left to stand and the ice melts.

Which row describes what happens as the ice is melting?

	temperature of mixture	energy change
A	increases	average kinetic energy of particles decreases
B	increases	energy is used to overcome attractive forces
C	stays the same	average kinetic energy of particles decreases
D	stays the same	energy is used to overcome attractive forces

- 3 Hydrogen chloride gas [M_r : HCl, 36.5] is released at P in the apparatus shown.

The universal indicator paper turns red after 38 s.



The experiment is repeated using sulfur dioxide gas [M_r : SO₂, 64].

What is the result for sulfur dioxide gas?

	universal indicator paper turns	time for universal indicator paper to change colour / s
A	blue	26
B	blue	51
C	red	26
D	red	51

4 Four statements about atoms are listed.

- 1 The centre of an atom is positively charged.
- 2 Protons and electrons are located in the nucleus.
- 3 Protons and electrons have the same mass.
- 4 Most of the mass of an atom is in the nucleus.

Which statements are correct?

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

5 The electronic configurations of two elements are given.

element L: 2,8,8,1

element M: 2,8,4

Which row identifies the group number and the period number for element L and element M?

	element L		element M	
	group number	period number	group number	period number
A	I	4	IV	3
B	I	4	III	4
C	IV	1	III	4
D	IV	1	IV	3

6 Which statement explains why isotopes of the same element have the same chemical properties?

- A** They have different numbers of protons in their nucleus.
- B** They have different numbers of neutrons in their nucleus.
- C** They have the same electronic configuration.
- D** They have the same number of electrons as protons.

7 Which statements about potassium chloride are correct?

- 1 It conducts electricity when solid because its ions are free to move.
- 2 It has a high melting point because it has strong intermolecular forces.
- 3 Its structure is a giant lattice of alternating positive and negative ions.
- 4 It is soluble in water.

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

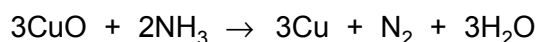
8 How many electrons are shared in **one** molecule of nitrogen and in **one** molecule of ethene?

	nitrogen	ethene
A	2	12
B	2	8
C	6	12
D	6	8

9 What is the total number of electrons in **one** molecule of ammonia, NH_3 ?

A 6 **B** 8 **C** 10 **D** 11

10 When heated, copper(II) oxide, CuO , reacts with ammonia, NH_3 .



8.5 g of ammonia reacts with an excess of copper(II) oxide to produce 26.4 g of copper.

What is the percentage yield of copper in this reaction?

A 27.5% **B** 32.2% **C** 55.0% **D** 82.5%

11 What is the empirical formula of ethanoic acid?

A CHO **B** CH_2O **C** $\text{C}_2\text{H}_2\text{O}$ **D** $\text{C}_2\text{H}_4\text{O}_2$

- 12 Magnesium chloride, MgCl_2 , contains magnesium ions and chloride ions.

How many chloride ions are present in **two** moles of magnesium chloride?

- A 6.02×10^{23}
 B 1.204×10^{24}
 C 2.408×10^{24}
 D 3.612×10^{24}

- 13 A metal object is electroplated with copper.

One electrode is the metal object and the other electrode is copper. The electrolyte is aqueous copper(II) sulfate.

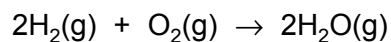
Which row shows the ionic half-equation for the reaction at the anode and the observation of the electrolyte?

	anode	electrolyte
A	$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	blue colour fades
B	$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	blue colour does not change
C	$\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$	blue colour fades
D	$\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$	blue colour does not change

- 14 Which statement about electrolysis is correct?

- A Chemical energy is converted to electrical energy.
 B Electrons flow through the electrolyte.
 C Ionic compounds are broken down.
 D Metals are formed at the positive electrode.

- 15 The reaction between hydrogen and oxygen releases 486 kJ/mol of energy.

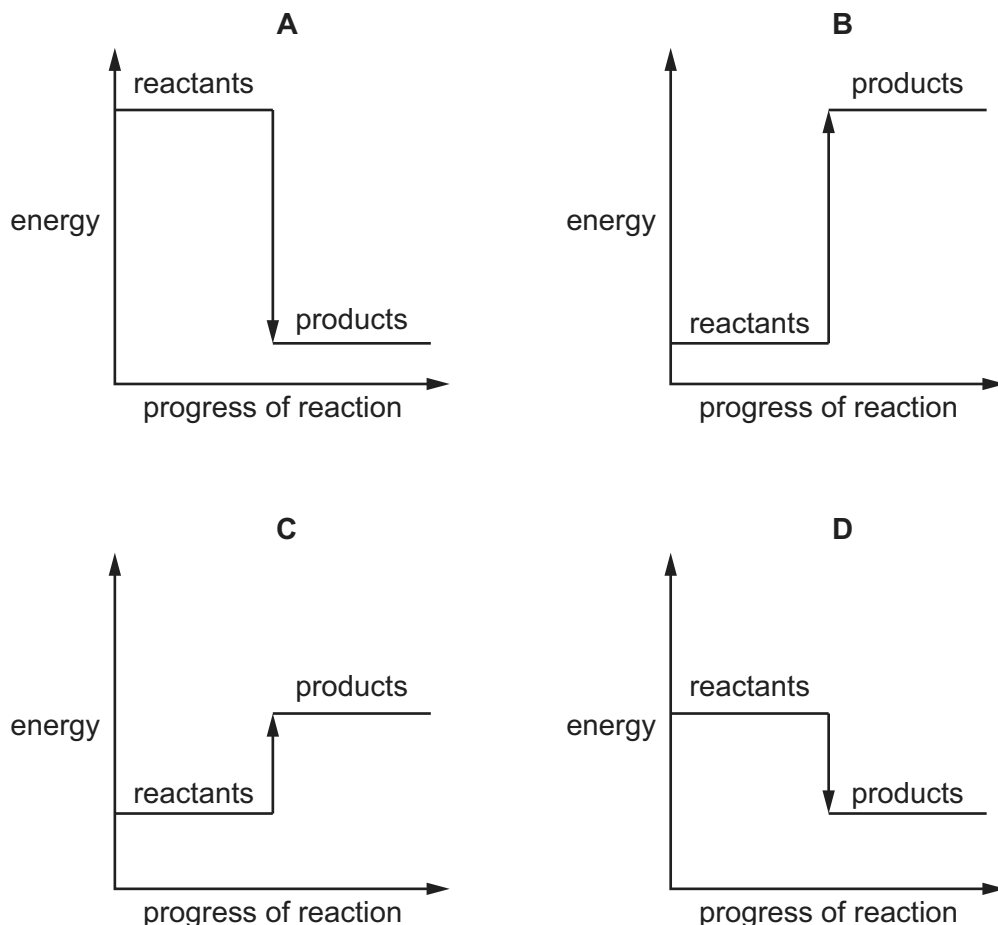


The bond energy of H–H is 436 kJ/mol and that of H–O is 464 kJ/mol.

What is the bond energy of O=O?

- A 430 kJ/mol
 B 458 kJ/mol
 C 498 kJ/mol
 D 984 kJ/mol
- 16 Which reaction pathway diagram shows the reaction that will give out the most energy?

The scale on the y-axis is the same in each diagram.



- 17 When calcium carbonate is heated strongly, a gas is given off.

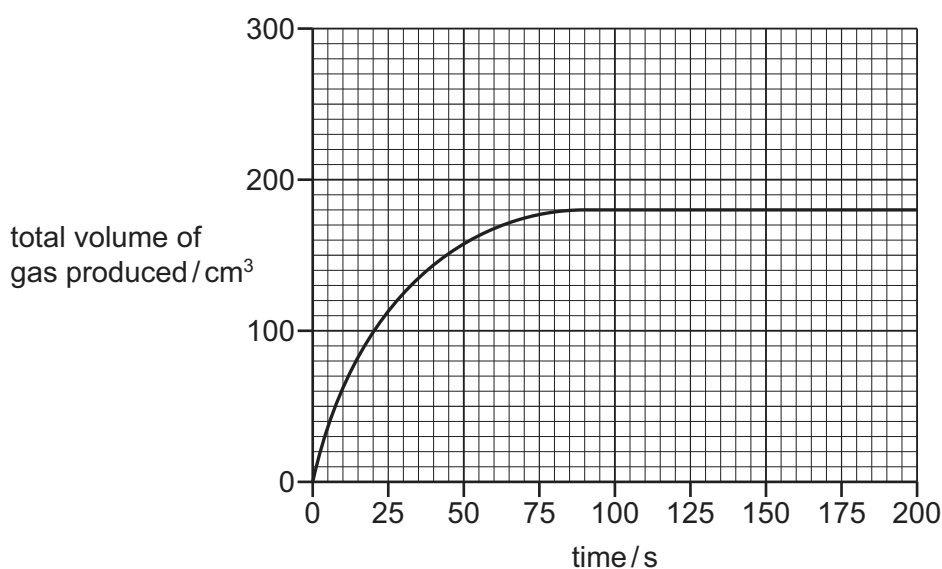
Which word describes this change?

- A chemical
- B exothermic
- C physical
- D reduction

- 18 Powdered magnesium carbonate is added to excess dilute hydrochloric acid.

The total volume of gas produced is measured over time.

A graph of the results is shown.



The experiment is repeated but the concentration of the hydrochloric acid is doubled.

All other conditions are kept the same.

Which statements about the second experiment are correct?

- 1 The final volume of gas is 360 cm^3 .
- 2 The reaction finishes before 90 seconds.
- 3 The activation energy of the reaction is lower.

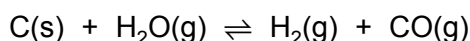
- A 1 and 2 B 1 and 3 C 2 and 3 D 2 only

19 Which statements explain why increasing the temperature changes the rate of a chemical reaction?

- 1 It increases the activation energy.
- 2 It increases the frequency of collisions between the reacting particles.
- 3 It increases the kinetic energy of the reacting particles.
- 4 It increases the number of particles per unit volume.

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

20 Hydrogen is made by reacting carbon with steam. The equation for the reaction is shown.



The forward reaction is endothermic.

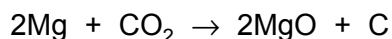
Which row describes changes in the pressure and the temperature that will **both** shift the position of equilibrium to the right?

	pressure	temperature
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

21 Which row shows the conditions used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process?

	pressure / atm	temperature / °C	catalyst
A	250	200	vanadium(V) oxide
B	2	450	vanadium(V) oxide
C	250	200	iron
D	2	450	iron

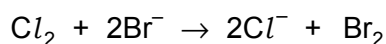
- 22 The equation for the reaction of magnesium with carbon dioxide is shown.



Which statement about this reaction is correct?

- A Magnesium is oxidised and carbon dioxide is reduced.
 - B Magnesium is reduced and carbon dioxide is oxidised.
 - C Magnesium and carbon dioxide are both oxidised.
 - D Magnesium and carbon dioxide are neither oxidised nor reduced.
- 23 Chlorine displaces bromine from aqueous potassium bromide.

The ionic equation for the reaction is shown.



Which statement about this reaction is correct?

- A Bromide ions act as an oxidising agent.
 - B Bromide ions are oxidised as electrons are lost.
 - C Chlorine acts as a reducing agent.
 - D Chlorine is reduced as electrons are lost.
- 24 Which gas is produced when ammonium chloride is warmed with aqueous sodium hydroxide?
- A ammonia
 - B chlorine
 - C hydrogen
 - D nitrogen
- 25 Which equation represents a solution of ethanoic acid in water?
- A $\text{HCOOH}(\text{aq}) \rightleftharpoons \text{HCOO}^-(\text{aq}) + \text{H}^+(\text{aq})$
 - B $\text{HCOOH}(\text{aq}) \rightarrow \text{HCOO}^-(\text{aq}) + \text{H}^+(\text{aq})$
 - C $\text{CH}_3\text{COOH}(\text{aq}) \rightleftharpoons \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}^+(\text{aq})$
 - D $\text{CH}_3\text{COOH}(\text{aq}) \rightarrow \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}^+(\text{aq})$

26 Four statements about the reactions of oxides with dilute hydrochloric acid and with aqueous sodium hydroxide are listed.

- 1 Aluminium oxide reacts with both dilute hydrochloric acid and aqueous sodium hydroxide.
- 2 Calcium oxide reacts with both dilute hydrochloric acid and aqueous sodium hydroxide.
- 3 Copper(II) oxide reacts with dilute hydrochloric acid but **not** with aqueous sodium hydroxide.
- 4 Sulfur dioxide does **not** react with either dilute hydrochloric acid or aqueous sodium hydroxide.

Which statements are correct?

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

27 Which statement about elements in Period 3 of the Periodic Table is correct?

- A** Aluminium is a non-metal in Group III.
B Argon is in Group VIII and has eight electrons in its outer electron shell.
C Magnesium is in Group II and has three electrons in its outer electron shell.
D Sulfur is a metal in Group VI.

28 Which row describes the structure of Group VII elements and the trend in their reactivity down the group?

	structure	reactivity down Group VII
A	diatomic	increases
B	diatomic	decreases
C	monatomic	increases
D	monatomic	decreases

29 Some information about four elements, P, Q, R and S, is shown.

	melting point in °C	density in g/cm ³	colour of chloride
P	1247	7.43	pink
Q	1410	2.33	white
R	1910	6.11	purple
S	115	2.07	red

Which elements are transition elements?

- A** P and R **B** P and S **C** Q and R **D** R and S

30 Propanoic acid is a carboxylic acid. It has similar chemical properties to ethanoic acid.

Which statements are correct?

- 1 Aqueous propanoic acid is a weaker acid than dilute hydrochloric acid.
- 2 Propanoic acid partially ionises in aqueous solution.
- 3 Propanoic acid reacts with ethanol to form propyl ethanoate.

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

31 Iron rusts in the presence of oxygen and water.

Which statements about the rusting of iron are correct?

- 1 Anhydrous iron(II) oxide is produced when iron rusts.
- 2 Iron rusts more quickly when attached to a piece of zinc.
- 3 Coating the iron with plastic prevents the iron from rusting.
- 4 Iron loses electrons when it rusts.

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

- 32** An iron nail is added to aqueous copper(II) sulfate and a different iron nail is added to aqueous magnesium sulfate.

The results are shown.

experiment	result
iron nail in aqueous copper(II) sulfate	nail is coated with a brown solid
iron nail in aqueous magnesium sulfate	no reaction

Which statement is correct?

- A** Copper atoms are oxidised more easily than magnesium atoms.
 - B** Copper atoms are reduced more easily than iron ions.
 - C** Iron atoms are oxidised more easily than copper atoms.
 - D** Iron atoms are reduced more easily than copper ions.
- 33** Which pollutant leads to the deoxygenation of water in ponds and lakes?
- A** fertilisers containing nitrates and phosphates
 - B** toxic metal compounds
 - C** combustion products of fossil fuels
 - D** acid rain
- 34** Which statement identifies a sample of water as pure?
- A** It melts at room temperature.
 - B** It turns anhydrous copper(II) sulfate blue.
 - C** It turns hydrated cobalt(II) chloride from blue to pink.
 - D** It boils at 100 °C.

35 Oxides of nitrogen are produced by car engines.

In a catalytic converter oxides of nitrogen are removed by reacting them with compound X.

Which row describes the type of reaction oxides of nitrogen undergo and identifies compound X?

	type of reaction	compound X
A	oxidation	carbon dioxide
B	oxidation	carbon monoxide
C	reduction	carbon dioxide
D	reduction	carbon monoxide

36 What is a disadvantage of producing ethanol using the catalytic addition of steam to ethene?

- A** the energy cost is low
- B** the process is continuous
- C** the process uses a non-renewable raw material
- D** the ethanol is pure

37 Which statement about the polymer PET is correct?

- A** It can be broken down into its monomers and re-polymerised.
- B** It is an addition polymer.
- C** It is a polyamide.
- D** It is made from amino acid monomers.

38 The formulae of five compounds are listed.

- 1 C_4H_{10}
- 2 $\text{C}_2\text{H}_5\text{OH}$
- 3 $\text{C}_4\text{H}_9\text{OH}$
- 4 $\text{C}_4\text{H}_9\text{COOH}$
- 5 $\text{C}_5\text{H}_{11}\text{OH}$

Which compounds are in the same homologous series?

- A** 1, 3 and 4
B 2, 3 and 5
C 3 and 4 only
D 3 and 5 only

39 Propane reacts with chlorine.

Which statements about this reaction are correct?

- 1 Ultraviolet light is used to provide the activation energy.
- 2 Propane undergoes an addition reaction.
- 3 One of the products is $\text{CH}_3\text{CH}_2\text{Cl}$.
- 4 One of the products is HCl .

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

40 Which statement about chromatography is correct?

- A** It is **not** possible for two different substances to have the same R_f value.
B It is only possible to use chromatography on substances which have a colour.
C It is possible to use chromatography on colourless substances using a locating agent.
D The R_f value of a substance = $\frac{\text{the distance travelled by the solvent}}{\text{the distance travelled by the substance}}$

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The Periodic Table of Elements

Group																																																																																																																																								
I	II											III	IV	V	VI	VII	VIII																																																																																																																							
Key atomic number atomic symbol name relative atomic mass 3 Li lithium 7 4 Be beryllium 9 11 Na sodium 23 12 Mg magnesium 24 19 K potassium 39 20 Ca calcium 40 37 Rb rubidium 85 38 Sr strontium 88 55 Cs caesium 133 56 Ba barium 137 87 Fr francium — 88 Ra radium — 1 H hydrogen 1 2 He helium 4 5 B boron 11 6 C carbon 12 13 Al aluminium 27 14 Si silicon 28 15 P phosphorus 31 16 S sulfur 32 17 Cl chlorine 35.5 18 Ar argon 40 31 Ga gallium 70 32 Ge germanium 73 33 As arsenic 75 34 Se selenium 79 35 Br bromine 80 36 Kr krypton 84 51 Sb antimony 122 52 Te tellurium 128 83 Bi bismuth 209 84 Po polonium — 115 Mc moscovium — 116 Lv livermorium — 117 Ts tennessine — 118 Og oganesson — 71 Lu lutetium 175 70 Yb ytterbium 173 103 Lr lawrencium — 102 No nobelium — lanthanoids <table><tr><td>57</td><td>La</td><td>lanthanum</td><td>139</td><td>58</td><td>Ce</td><td>cerium</td><td>140</td><td>59</td><td>Pr</td><td>praseodymium</td><td>141</td><td>60</td><td>Nd</td><td>neodymium</td><td>144</td><td>61</td><td>Pm</td><td>promethium</td><td>—</td><td>62</td><td>Sm</td><td>samarium</td><td>150</td><td>63</td><td>Eu</td><td>europium</td><td>152</td><td>64</td><td>Gd</td><td>gadolinium</td><td>157</td><td>65</td><td>Tb</td><td>terbium</td><td>159</td><td>66</td><td>Dy</td><td>dysprosium</td><td>163</td><td>67</td><td>Ho</td><td>holmium</td><td>165</td><td>68</td><td>Er</td><td>erbium</td><td>167</td><td>69</td><td>Tm</td><td>thulium</td><td>169</td><td>70</td><td>Yb</td><td>ytterbium</td><td>173</td><td>71</td><td>Lu</td><td>lutetium</td><td>175</td></tr></table> actinoids <table><tr><td>89</td><td>Ac</td><td>actinium</td><td>—</td><td>90</td><td>Th</td><td>thorium</td><td>232</td><td>91</td><td>Pa</td><td>protactinium</td><td>231</td><td>92</td><td>U</td><td>uranium</td><td>238</td><td>93</td><td>Np</td><td>neptunium</td><td>—</td><td>94</td><td>Pu</td><td>plutonium</td><td>—</td><td>95</td><td>Am</td><td>americium</td><td>—</td><td>96</td><td>Cm</td><td>curium</td><td>—</td><td>97</td><td>Bk</td><td>berkelium</td><td>—</td><td>98</td><td>Cf</td><td>californium</td><td>—</td><td>99</td><td>Es</td><td>einsteinium</td><td>—</td><td>100</td><td>Fm</td><td>fermium</td><td>—</td><td>101</td><td>Md</td><td>mendelevium</td><td>—</td><td>102</td><td>No</td><td>nobelium</td><td>—</td><td>103</td><td>Lr</td><td>lawrencium</td><td>—</td></tr></table> The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).																	57	La	lanthanum	139	58	Ce	cerium	140	59	Pr	praseodymium	141	60	Nd	neodymium	144	61	Pm	promethium	—	62	Sm	samarium	150	63	Eu	europium	152	64	Gd	gadolinium	157	65	Tb	terbium	159	66	Dy	dysprosium	163	67	Ho	holmium	165	68	Er	erbium	167	69	Tm	thulium	169	70	Yb	ytterbium	173	71	Lu	lutetium	175	89	Ac	actinium	—	90	Th	thorium	232	91	Pa	protactinium	231	92	U	uranium	238	93	Np	neptunium	—	94	Pu	plutonium	—	95	Am	americium	—	96	Cm	curium	—	97	Bk	berkelium	—	98	Cf	californium	—	99	Es	einsteinium	—	100	Fm	fermium	—	101	Md	mendelevium	—	102	No	nobelium	—	103	Lr	lawrencium	—
57	La	lanthanum	139	58	Ce	cerium	140	59	Pr	praseodymium	141	60	Nd	neodymium	144	61	Pm	promethium	—	62	Sm	samarium	150	63	Eu	europium	152	64	Gd	gadolinium	157	65	Tb	terbium	159	66	Dy	dysprosium	163	67	Ho	holmium	165	68	Er	erbium	167	69	Tm	thulium	169	70	Yb	ytterbium	173	71	Lu	lutetium	175																																																																													
89	Ac	actinium	—	90	Th	thorium	232	91	Pa	protactinium	231	92	U	uranium	238	93	Np	neptunium	—	94	Pu	plutonium	—	95	Am	americium	—	96	Cm	curium	—	97	Bk	berkelium	—	98	Cf	californium	—	99	Es	einsteinium	—	100	Fm	fermium	—	101	Md	mendelevium	—	102	No	nobelium	—	103	Lr	lawrencium	—																																																																													